Henry C. Shires

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My Reflections on Engineering Ethics and My Role

In modern society, scientists and engineers are directly responsible for solving some of the world's biggest problems. As one can easily guess, these roles are not trivial, requiring diverse and complex skill sets and a passion for improving the world. While most engineers intend to positively impact any situation they find themselves in, humans are imperfect and sometimes fall short in instances that require complete focus and unbiased thinking. Additionally, people do not live in a perfect world, and as such, there comes time for difficult decisions, sacrifices, and compromises that can have negative implications no matter the option decided or route taken. Engineers are faced with these dilemmas, or "forced choices between two or more choices that are equally bad," as defined by the 5 P's of Ethics, and as such, engineers have designed a standard or ethics code to follow in case a dilemma is faced. Not only does this code provide a standardized set to follow, but it also enables engineers to constantly reflect on the impacts of their work and if they are "good."

Should a new feature be implemented into a design that could put a portion of the user base at risk to reduce manufacturing expenses? Should extra precautions be instated for sensitive data usage, even if tedious and time-consuming? For electrical and electronics engineers specifically, questions like these can be best answered by the IEEE Code of Ethics, a set of guidelines to consider when resolving a dilemma, involving accepting responsibility for actions, avoiding conflicts of interest, rejecting bribery, having integrity, seeking criticism, avoiding malicious or harmful actions, and assisting peers. The Code of Ethics also serves as a mechanism to hold members of the IEEE accountable for their actions as engineers, whether individually or on behalf of a company. At its core, this code can be paraphrased into the golden rule: "Treat others the way you want to be treated." All things should be engineered for the

safety and sustainability of others to improve people's lives rather than take from them, and this code creates a very tangible set of rules to validate or invalidate an engineer's practice with the golden rule.

In class, we discussed multiple real and hypothetical cases involving ethical dilemmas. We discussed how automakers Volkswagen and Ford had their share of poor decisions made to cut manufacturing costs. In 2015, Volkswagen (VW) attempted to deploy emissions test cheating software on nearly 11 million vehicles, with a design of vehicle engines causing them to exceed the US's maximum nitrogen oxide pollution law, prohibiting their sale in the country. Given the significant economic value the US market brings to automakers, and rather than going backward in the vehicle design process, VW decided to bypass the law by cheating EPA emissions tests with software hacks on existing vehicles. This lie greatly threatened public health and left Volkswagen with criminal allegations. In Ford's case, a design flaw in their 1972 Pinto vehicle left the gas tank vulnerable to puncture and explosion in certain accidents that would kill the occupants. Ford was aware of the flaw and sold the car anyway, and as predicted, at least one case of this occurring appeared in the news. Rather than saving money by ignoring a manufacturing error, they destroyed their reputation and paid for the costs of many lives lost.

Specifically for VW, this issue relates to numbers 1, 3, 7, and 9 of the IEEE Code of Ethics and numbers 1 and 5 from the 5 Ps. Additionally, VW lacked integrity, honesty, charity, and responsibility from the list of virtue ethics. As a class, we determined that VW did not make a decision consistent with the public's safety, health, and welfare as they chose to risk polluting the globe and profit from a design flaw (1). VW was not initially honest with testing organizations in the US when certifying their vehicle emissions status (3). VW waited for an extensive investigation before correcting any error in their behavior (7) and risked injuring others due to their malicious actions (9). These code of ethics violations resulted from priorities (5) within the organization, valuing money and time over public safety, thus placing pressure (1) on its employees to abide by this strict priority, cutting corners and violating laws whenever necessary. As the article excerpt states, "fitting 11m cars with a piece of software which artificially reduces vehicle emissions during regulators' tests is not the work of a few rouge employees.

That decision was taken and put into action by people of reasonable seniority." This pressure and profit-prioritizing culture likely reflected the values of VW executives and product managers, forcing their teams to comply with revenue goals and benchmarks. Integrity and honesty are related in this case, as the lying and cheating by VW are in direct contrast to these virtues. Additionally, it was irresponsible of VW to enable this lie on millions of their vehicles and exploitation of leadership from the executives on their employees. No charity existed within the organization due to the large exploitation of leadership to force employees to implement this solution into vehicles. An additional or new virtue would not apply to this case, as only one large ethical violation was committed (lying/cheating) across the entire company, with multiple actions described by the virtues already listed.

I was surprised that one of my classmates mentioned that other automakers had attempted similar cheats due to the extremely strict regulations. These regulations limit foreign manufactured costs, incentivizing companies to exploit legal loopholes to reduce costs to sell in the large US market. My perspective was changed slightly, as a law or other factor needs to be amended to reduce the incentive for companies to lie. While simply saying an ethical violation could solve the problem, if this cheating has been established as an industry norm, it may be time to call into question the laws in practice and if they are reasonable.

As a software startup co-founder, I frequently encounter many ethical situations with my team. How much should we charge for our product? How should we ensure the security and privacy of our customers' data? Should we undercut our competitors and then raise prices when we have a majority market share or keep our prices consistent? These are all questions my team has asked to ensure we remain profitable in addition to solving the problem we engineered a solution for. As CEO, I must ensure our mission, vision, and values are upheld in all decisions and work we accomplish, so we evaluate each question based on our own virtue ethics.

Will the decision positively or negatively affect our customers? Will the decision portray us as a loyal and trustworthy company that respects its customers? Do we care about outcomes other than just

profit? Ultimately, the right decision we make should indeed positively impact our customers and keep our customer's trust in us so we can continue to benefit from our market and make a difference with our product. More money can be made if corners are cut, but how far will cut corners drive a business if trust with customers is suddenly broken?

While codes of ethics and rules to follow surely help people remain on the right path when designing, building, and sharing ideas or solutions, it is important to remember that simply having a code does not prevent mistakes like Ford or Volkswagen (These codes clearly existed during both of these controversies). Those who actually take meaningful action to do what is right propel our world forward into improvement and change. Not everyone will make the right decisions 100% of the time, and not all dilemmas have a straightforward solution, but if engineers actively try to seek the right decision, the right impacts, and the good for *all* people, great things are sure to follow. Mistakes from the past or scenarios are simply opportunities to be inspired to learn from a lesson and do better than before. Thus, we should all be inspired, and I am, to stand up for the right thing in any situation, big or small, as it might make all the difference in the world.